Study design

1 Reduce

18 satellite and gridded modalities

3,077 counties

2003 to 2019

several summaries per field, one pass

2 Model

581 admitted columns

no socioeconomic or clinical input

XGBoost, 8 county-grouped folds

$R^2 = 0.666$

interventional SHAP

3 Compare

census of 47 fields

central vs non-central

SHAP vs mutual information

matched refit

8 instruments, nulls

4 What survives

atmospheric lifetime (13 species)

turning points: noise, 100th meridian

model error vs socioeconomic data